0704. Binary Search

Easy · Binary Search

leetcode.com/problems/binary-search

Problem

- Find target in a sorted array of distinct integers; return its index or -1, in O(log n).

Intuition

- Compare with the middle element; half of the remaining range can be discarded each step.

Brute Force

- Linear scan: O(n).

Optimal Approach

- Lo = 0, hi = n-1.
- Mid = lo + (hi - lo) / 2.
- Equal → return mid; a[mid] < target → lo = mid+1; else hi = mid-1.

Worked Example

```
nums = [-1,0,3,5,9,12], target = 9.

lo=0 hi=5 mid=2 (3)  < 9 -> lo=3
lo=3 hi=5 mid=4 (9)  found -> 4
```

Java Solution

```java
public class BinarySearch {
    public int search(int[] nums, int target) {
        int lo = 0, hi = nums.length - 1;
        while (lo <= hi) {
            int mid = lo + (hi - lo) / 2;
            if (nums[mid] == target) return mid;
            if (nums[mid] < target) lo = mid + 1;
            else hi = mid - 1;
        }
        return -1;
    }
}
```

Complexity

- Time: O(log n). Space: O(1).

Pitfalls

- (lo + hi) / 2 can overflow; use lo + (hi - lo) / 2.
- Loop condition lo <= hi for the closed interval [lo, hi].